Fiscal Year:	FY 2008	**Task Last Updated:**	FY 03/18/2008
PI Name:	Delp, Michael Ph.D.		
Project Title:	Arterial Remodeling and Functional Adaptations Induced by Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	None		
PI Email:	mdelp@fsu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	352-214-1195
Organization Name:	Florida State University		
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PI Web Page:			
City:	Tallahassee	**State:**	FL
Zip Code:	32306-0001	**Congressional District:**	2
Comments:	Previous affiliations were University of Florida (mid-2007-June 2014), West Virginia University (mid-2005 to mid-2007), and Texas A&M University (1995 to mid-2005).		
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	09/01/2006	**End Date:**	12/31/2007
No. of Post Docs:	3	**No. of PhD Degrees:**	8
No. of PhD Candidates:	8	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle		
Flight Assignment:	STS-107 NOTE: PI moved to University of Florida from WVU in mid-2007 (12/07) NOTE: Received NCE through 12/31/2007 (original end date was 12/31/2006), per A. Chu (1/07) NOTE: PI moved to West Va. U from TX A&M in Sept 2005 (per A. Chu 7/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA06CB60G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Tail suspension or hindlimb unloading of rats has been used as a ground-based model of microgravity because tail suspension, like actual microgravity, 1) induces a headward shift of fluids and 2) diminishes the contractile activity of hindlimb skeletal muscles that are normally involved in maintaining posture. It is now evident that the shift of fluid toward the head and the unloading of postural muscles alter the mechanical forces exerted on arteries, the vessels responsible for regulating blood flow and arterial blood pressure. For example, the smooth muscle cells in the small arteries of hindlimb skeletal muscle atrophies with tail suspension, and results in a diminished ability of these vessels to constrict. The compromised ability to constrict reduces the ability of the rats to maintain normal blood pressure, and consequently, these animals are susceptible to orthostatic hypotension (an abnormal reduction in arterial pressure when the body position of the rat is moved from a horizontal position to a 70o vertical head-up position). The purpose of the present study is to determine whether the fluid shifts and reduced activity of postural muscles that are induced by microgravity similarly alter rodent arterial vessel structure and function. In other words, does space flight result in a diminished ability of arteries to constrict, and if so, is this the result of smooth muscle atrophy. This work will provide potentially important information about the mechanisms underlying the reduced fitness and the inability to properly regulate blood pressure of astronauts returning to Earth. The purpose of the present study is to determine whether the fluid shifts and reduced activity of postural muscles that are induced by microgravity similarly alter rodent arterial vessel structure and function. In other words, does space flight result in a diminished ability of arteries to constrict, and if so, is this the result of smooth muscle atrophy. This work will provide potentially important information about the mechanisms underlying the reduced fitness and the inability to properly regulate blood pressure of astronauts returning to Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Orthostatic intolerance is a common health problem for Americans, particularly the elderly. Therefore, this work will provide potentially important information about the mechanisms underlying orthostatic intolerance and the reduced fitness as it pertains to vascular biology on Earth.
Task Progress:	This study demonstrated the feasibility of successfully transporting a considerable number of microscopes and other technical equipment to the KSC laboratory and coordinating a large group of people (21 investigators, postdocs and graduate students) in order to conduct physiological studies of small isolated blood vessels. No conclusions or final investigation results due to Columbia Shuttle accident. However, ground-based studies were conducted to determine the effects of simulated microgravity on the splanchnic circulation and the effects of aging on skeletal muscle and bone circulations, each of which are affected by space flight in humans. Note: publications listed below cover the entire grant period, from March 2000 to December 2007.
Bibliography Type:	Description: (Last Updated: 07/09/2025)
Articles in Peer-reviewed Journals	Lesniewske LA, Donato AJ, Behnke BJ, Woodman CR, Laughlin MH, Ray CA, Delp MD. "Decreased NO signaling leads to enhanced vasoconstrictor responsiveness in skeletal muscle arterioles of the ZDF rat prior to overt diabetes and hypertension." Am J Physiol Heart Circ Physiol. 2008 Apr;294(4):H1840-50. Epub 2008 Feb 1. http://dx.doi.org/10.1152/ajpheart.00692.2007 ; PMID: 18245568 , Apr-2008
Articles in Peer-reviewed Journals	Behnke BJ, Zawieja DC, Gashev AA, Ray CA, Delp MD. "Diminished mesenteric vaso- and venoconstriction and elevated plasma ANP and BNP with simulated microgravity." J Appl Physiol. 2008 May;104(5):1273-80. PMID: 18218919 , May-2008
Articles in Peer-reviewed Journals	Delp MD, Behnke BJ, Spier SA, Wu G, Muller-Delp JM. "Ageing diminishes endothelium-dependent vasodilatation and tetrahydrobiopterin content in rat skeletal muscle arterioles." J Physiol. 2008 Feb 15;586(4):1161-8. PMID: 18063659 , Feb-2008
Articles in Peer-reviewed Journals	Ramsey MW, Behnke BJ, Prisby RD, Delp MD. "Effects of aging on adipose resistance artery vasoconstriction: possible implications for orthostatic blood pressure regulation." J Appl Physiol. 2007 Nov;103(5):1636-43. PMID: 17885023 , Nov-2007
Articles in Peer-reviewed Journals	Prisby RD, Ramsey MW, Behnke BJ, Dominguez JM 2nd, Donato AJ, Allen MR, Delp MD. "Aging reduces skeletal blood flow, endothelium-dependent vasodilation, and NO bioavailability in rats." J Bone Miner Res. 2007 Aug;22(8):1280-8. PMID: 17451371 , Aug-2007
Articles in Peer-reviewed Journals	Donato AJ, Lesniewski LA, Delp MD. "Ageing and exercise training alter adrenergic vasomotor responses of rat skeletal muscle arterioles." J Physiol. 2007 Feb 15;579(Pt 1):115-25. PMID: 17082231 , Feb-2007
Articles in Peer-reviewed Journals	Behnke BJ, Prisby RD, Lesniewski LA, Donato AJ, Olin HM, Delp MD. "Influence of ageing and physical activity on vascular morphology in rat skeletal muscle." J Physiol. 2006 Sep 1;575(Pt 2):617-26. PMID: 16644803 , Sep-2006
Articles in Peer-reviewed Journals	Prisby RD, Wilkerson MK, Sokoya EM, Bryan RM Jr, Wilson E, Delp MD. "Endothelium-dependent vasodilation of cerebral arteries is altered with simulated microgravity through nitric oxide synthase and EDHF mechanisms." J Appl Physiol. 2006 Jul;101(1):348-53. PMID: 16627679 , Jul-2006
Articles in Peer-reviewed Journals	Gashev AA, Delp MD, Zawieja DC. "Inhibition of active lymph pump by simulated microgravity in rats." Am J Physiol Heart Circ Physiol. 2006 Jun;290(6):H2295-308. PMID: 16399874 , Jun-2006
Articles in Peer-reviewed Journals	Wilkerson MK, Lesniewski LA, Golding EM, Bryan RM Jr, Amin A, Wilson E, Delp MD. "Simulated microgravity enhances cerebral artery vasoconstriction and vascular resistance through endothelial nitric oxide mechanism." Am J Physiol Heart Circ Physiol. 2005 Apr;288(4):H1652-61. PMID: 15576439 , Apr-2005

Articles in Peer-reviewed Journals	Donato AJ, Lesniewski LA, Delp MD. "The effects of aging and exercise training on endothelin-1 vasoconstrictor responses in rat skeletal muscle arterioles." Cardiovasc Res. 2005 May 1;66(2):393-401. PMID: 15820208 , May-2005
Articles in Peer-reviewed Journals	Spier SA, Delp MD, Meininger CJ, Donato AJ, Ramsey MW, Muller-Delp JM. "Effects of ageing and exercise training on endothelium-dependent vasodilatation and structure of rat skeletal muscle arterioles." J Physiol. 2004 May 1;556(Pt 3):947-58. PMID: 15004211 , May-2004
Articles in Peer-reviewed Journals	Papadopoulos A, Delp MD. "Effects of hindlimb unweighting on the mechanical and structure properties of the rat abdominal aorta." J Appl Physiol. 2003 Feb;94(2):439-45. PMID: 12391084 , Feb-2003
Articles in Peer-reviewed Journals	Colleran PN, Behnke BJ, Wilkerson MK, Donato AJ, Delp MD. "Simulated microgravity alters rat mesenteric artery vasoconstrictor dynamics through an intracellular Ca2+ release mechanism." Am J Physiol Regul Integr Comp Physiol. 2008 May;294(5):R1577-85. Epub 2008 Mar 19. PMID: 18353882 , May-2008
Articles in Peer-reviewed Journals	Muller-Delp JM, Spier SA, Ramsey MW, Delp MD. "Aging impairs endothelium-dependent vasodilation in rat skeletal muscle arterioles." Am J Physiol Heart Circ Physiol. 2002 Oct;283(4):H1662-72. PMID: 12234821 , Jan-2003
Articles in Peer-reviewed Journals	Muller-Delp J, Spier SA, Ramsey MW, Lesniewski LA, Papadopoulos A, Humphrey JD, Delp MD. "Effects of aging on vasoconstrictor and mechanical properties of rat skeletal muscle arterioles." Am J Physiol Heart Circ Physiol. 2002 May;282(5):H1843-54. PMID: 11959651 , Jan-2003
Articles in Peer-reviewed Journals	Bloomfield SA, Hogan HA, Delp MD. "Decreases in bone blood flow and bone material properties in aging Fischer-344 rats." Clin Orthop Relat Res. 2002 Mar;(396):248-57. PMID: 11859250 , Jan-2003
Articles in Peer-reviewed Journals	Wilkerson MK, Colleran PN, Delp MD. "Acute and chronic head-down tail suspension diminishes cerebral perfusion in rats." Am J Physiol Heart Circ Physiol. 2002 Jan;282(1):H328-34. PMID: 11748078 , Jan-2003
Articles in Peer-reviewed Journals	Ray CA, Vasques M, Miller TA, Wilkerson MK, Delp MD. "Effect of short-term microgravity and long-term hindlimb unloading on rat cardiac mass and function." J Appl Physiol. 2001 Sep;91(3):1207-13. PMID: 11509517 , Jan-2003
Articles in Peer-reviewed Journals	Behnke BJ, Ramsey MW, Stabley JN, Dominguez JM 2nd, Davis RT 3rd, McCullough DJ, Muller-Delp JM, Delp MD. "Effects of aging and exercise training on skeletal muscle blood flow and resistance artery morphology." J Appl Physiol. 2012 Dec 1;113(11):1699-708. http://dx.doi.org/10.1152/japplphysiol.01025.2012 ; PubMed PMID: 23042906 , Dec-2012
Articles in Peer-reviewed Journals	Davis RT 3rd, Stabley JN, Dominguez JM 2nd, Ramsey MW, McCullough DJ, Lesniewski LA, Delp MD, Behnke BJ. "Differential effects of aging and exercise on intra-abdominal adipose arteriolar function and blood flow regulation." J Appl Physiol. 2013 Mar 15;114(6):808-15. http://dx.doi.org/10.1152/japplphysiol.01358.2012 ; PubMed PMID: 23349454 , Mar-2013
Books/Book Chapters	Delp MD. "Microgravity-induced orthostatic intolerance: An arterial microvascular mechanism." in "Adaptation Biology and Medicine : Vol. 3. New Frontiers." Ed. J. Moravec, N. Takeda, and P.K. Singal. New Delhi, India : Narosa Publishing House, p. 144-155, 2002. , Jan-2003
Journal/Magazine covers	Prisby RD, Ramsey MW, Behnke BJ, Dominguez JM 2nd, Donato AJ, Allen MR, Delp MD. "Cover in Journal of Bone Mineral Research for article, “Aging reduces skeletal blood flow, endothelium-dependent vasodilation, and NO bioavailability in rats." J Bone Miner Res. 2007 Aug;22(8):1280-8. PMID: 17451371 , Aug-2007